

E-Banking — Detailed Lecture Notes

JKSSB Computer Awareness

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39.1 What e-banking covers

Electronic banking, or e-banking, means using banking services without visiting the branch. The account itself stays in the bank; only the channel used to reach it changes. **Net banking** is banking through the bank's website in a browser. **Mobile banking** is banking through the bank's app on the phone. **UPI** (Unified Payments Interface) apps move money directly between bank accounts from the phone. An ATM, the net-banking site, the mobile app, and a UPI app are therefore four doors into the same account.

Definition (E-banking). *The use of banking services through electronic channels — net banking, mobile banking, UPI, and ATMs — instead of visiting the branch in person.*

Term	Meaning in this lesson
Net banking	Banking through the bank's website in a browser.
Mobile banking	Banking through the bank's official app.
UPI	Unified Payments Interface; instant phone-based payments between accounts.
VPA	Virtual Payment Address (for example name@bank); identifies the account without the account number.
ATM PIN	Personal Identification Number used with the physical card.
UPI PIN	Personal Identification Number authorising UPI payments.
CVV	Card Verification Value printed on the card.
OTP	One-Time Password sent by SMS, valid for one transaction only.
MFA	Multi-Factor Authentication; checking two or more factors together.

The rest of this lesson uses these terms in one consistent sense. A **PIN** is a secret typed by the holder. **CVV** is printed on the card. **OTP** arrives by SMS and expires. **MFA** is the combination of two or more different proofs of identity.

39.2 UPI in detail

UPI stands for Unified Payments Interface. Each user is identified by a **VPA** (Virtual Payment Address), such as name@bank, so money can be sent without sharing the account number or IFSC. Transfers are **instant** and work 24×7 , including Sundays and holidays. Every payment is authorised on the payer's own phone by entering the **UPI PIN**, which is set separately for each bank account inside the UPI app. The central rule of UPI safety is short: entering the UPI PIN always sends money, and receiving money never needs any PIN.

Other electronic transfer modes appear in nearby options and should be kept in one line each. **IMPS** (Immediate Payment Service) is an instant bank-to-bank transfer available 24×7 . **NEFT** (National Electronic Funds Transfer) settles in batches and has no minimum amount. **RTGS** (Real Time Gross Settlement) settles large-value transfers instantly and carries a minimum amount. UPI and IMPS are the instant, phone-friendly modes; NEFT is batched; RTGS is for large values.

Definition (UPI). *Unified Payments Interface: an instant, round-the-clock system for moving money between bank accounts from a phone, identified by VPA and authorised by a UPI PIN.*

39.3 The four secret codes

Four codes look alike and are deliberately mixed in distractors. Each lives in a different place and does a different job.

ATM PIN. The Personal Identification Number used with the physical card at an ATM or shop terminal. It is memorised, typed on the machine, and authorises card transactions on that channel. It never authorises a UPI payment.

UPI PIN. The Personal Identification Number typed in the phone to authorise a UPI payment. It is set per bank account in the UPI app and can differ from the ATM PIN. Approving any screen with the UPI PIN sends money.

CVV. The Card Verification Value: the 3-digit number printed on the back of the card (4 digits on the front for some cards). It is entered with the card number and expiry date for online card payments and proves the card is in hand. It is fixed and reusable for the life of the card.

OTP. The One-Time Password: a short numeric code sent by SMS that is valid for one transaction only and for a few minutes. It is generated fresh each time and expires after use or after the time window.

Code	Where it lives	How long it lasts	Typical use
ATM PIN	In memory	Until changed	Card at ATM or shop terminal
UPI PIN	In memory	Until changed	Authorising a UPI payment
CVV	Printed on card	Life of the card	Online card payment with card number
OTP	Sent by SMS	One transaction, minutes	Second approval for a login or payment

Example 39.3.1. *A question states, “The 3-digit number on the back of the card was sent by SMS and expired after one use.” This mixes two codes. The number printed on the back of the card is the **CVV**, which is fixed and reusable; the code that arrives by SMS and expires after one use is the **OTP**. The stem has attached OTP behaviour to the CVV name.*

39.4 Authentication and multi-factor authentication

Authentication means proving “I am the account holder” before the bank acts. It uses three kinds of factor. **Something you know** is a password or PIN. **Something you have** is the phone, the card, or the OTP that arrives on the registered mobile number. **Something you are** is a fingerprint or face scan.

Definition (Multi-Factor Authentication (MFA)). *Authentication that checks two or more different factors together, for example a password (know) plus an SMS OTP (have), or a PIN plus a fingerprint.*

A login password plus an SMS OTP is therefore **two-factor** authentication, not one: the password proves knowledge and the OTP proves possession of the registered phone. Asking for the OTP a second time does not add a new factor; a fingerprint or card would. The exam trap is to call a password-plus-OTP login “single-factor” or to call two passwords “multi-factor”; factors must be of *different* kinds.

39.5 QR codes and collect requests

A **QR** (Quick Response) code is the square barcode the merchant displays. Scanning it with the UPI app fills in the merchant’s payment address; the payer then checks the **merchant name and amount** on their own screen and approves with the UPI PIN. A QR never asks for a PIN or OTP by itself. A common fraud is a fake QR sticker pasted over the real one, so the displayed name on the payer’s screen is the check that matters: if the name does not match the shop, stop.

A **collect request** is a message asking the receiver to pay: it appears as “pay Rs. X to Y, approve with UPI PIN”. Approving it and entering the UPI PIN sends the receiver’s money to the requester. No refund, cashback, prize, or buyer ever needs the seller to enter a UPI PIN; entering the PIN always means paying. Unexpected collect requests, especially ones with remarks like “refund” or “cashback”, must be declined or ignored. To receive money, the receiver only shares a VPA or QR and enters nothing.

Example 39.5.1. *A buyer sends a Rs. 500 collect request to a seller with the remark “advance, approve to receive”. The seller’s screen shows “pay Rs. 500”, and approving with the UPI PIN would send Rs. 500 to the buyer. The remark is false: receiving needs no PIN. The safe action is to decline and ask the buyer to send the payment instead.*

39.6 Safe-banking rules

The rules below are tested almost verbatim, so they are stated exactly.

1. The bank **never** asks for OTP, CVV, ATM PIN, or UPI PIN by call, SMS, email, or link. Any such request is fraud.
2. **Never share** an OTP for any reason — to “cancel”, to “verify”, or to “receive” a payment. Sharing it completes the transaction the fraudster started.
3. Do not open **KYC-update or account-blocked** links from SMS. Open the bank’s site or app by typing the address or using the official app.
4. **Never install** a screen-sharing or remote-access app at a caller’s request; it exposes OTPs, PINs, and balances on the screen.
5. Use **typed addresses and official apps**, and check the **padlock (https)** on the banking site before logging in.
6. Avoid **public Wi-Fi** for banking; prefer mobile data or a trusted network.
7. **Log out** after net banking, lock the UPI app with the device lock, and keep **SMS and email alerts on** so an unknown debit is noticed at once.

39.7 Exam retrieval and common confusions

Six distinctions prevent most errors on this topic.

1. ATM PIN is for the **card** channel; UPI PIN is for the **UPI** channel. They are set and used separately.
2. CVV is **printed on the card** and reusable; OTP is **sent by SMS** for one transaction and expires.
3. Entering the UPI PIN **always pays**; receiving money needs **no PIN**.
4. A password plus an SMS OTP is **two-factor** authentication (know + have); two passwords are still one factor.
5. The bank **never asks** for OTP, CVV, or any PIN. Any caller who asks is not the bank.
6. A QR is verified by the **name and amount on your own screen**; a pasted-over sticker is defeated by that check.

Example 39.7.1. *An item asks what is needed to receive a UPI refund. Trace the rule: receiving needs a VPA or QR and no PIN; any screen asking for the UPI PIN is a payment screen. The answer is that no PIN or OTP is entered to receive; the “approve to receive” request is a collect-request fraud.*

Exercises

1. Expand UPI, OTP, CVV, PIN, QR, and MFA.
2. Distinguish the ATM PIN from the UPI PIN: where is each used?
3. Distinguish CVV from OTP: where does each live, and how long does each last?
4. Explain why a login password plus an SMS OTP counts as two-factor authentication.
5. A seller receives a collect request marked “refund, approve to receive”. What should the seller do, and why?

Solutions

1. UPI — Unified Payments Interface; OTP — One-Time Password; CVV — Card Verification Value; PIN — Personal Identification Number; QR — Quick Response code; MFA — Multi-Factor Authentication.
2. The ATM PIN is typed on a machine with the physical card at an ATM or shop terminal. The UPI PIN is typed in the phone to authorise a UPI payment, set per bank account in the UPI app. Neither works on the other’s channel.
3. CVV is printed on the card and is reusable for the life of the card; it is entered with the card number for online card payments. OTP is sent by SMS, is valid for one transaction only and a few minutes, and then expires.
4. The password is something the user knows and the OTP on the registered phone is something the user has. Two different factor kinds are checked, so the login is two-factor (a form of MFA).
5. Decline or ignore it. A collect request approved with the UPI PIN sends the seller’s money; receiving a refund needs no PIN. The remark is a fraud pattern.